

HOME IMPROVEMENT

The home improvement *math cheat sheet*

The formulas behind CalcForged's paint, flooring, concrete, and mulch calculators — with worked examples you can check in the store aisle. Planning estimates, not gospel: your surface, product label, and local prices get the final word.

01 Paint: buy by wall area, not by room size

STEP	RULE	12 × 10 ROOM, 8 FT WALLS
Wall area	perimeter × height – 21 sq ft per door – 12 sq ft per window	≈ 307 sq ft
Paint needed	area × coats ÷ 375 sq ft per gallon	≈ 1.6 gal (2 coats)
Buy	round up to whole gallons	2 gallons

SHEEN GUIDE

Eggshell for most living spaces · **Satin** for high-traffic and moisture (kitchens, baths) · **Flat** for ceilings. Two coats is the default for a consistent finish.

02 Flooring: order with waste, buy whole boxes

STEP	RULE	120 SQ FT ROOM
Room area	length × width (split L-shaped rooms into rectangles)	120 sq ft
Waste	+10% straight lay · +15% diagonal or herringbone	132 sq ft to order
Boxes	order area ÷ box coverage, rounded up	7 boxes (20 sq ft/box)

KEEP EXTRAS

Leftover planks are future repair stock. Write the color/lot number on the spare box and store it flat and dry.

03 Concrete: sold by volume, ordered by the yard

STEP	RULE	12 × 10 SLAB, 4 IN THICK
Base volume	slab: $L \times W \times (\text{thickness in} \div 12)$ · column: $\pi \times \text{radius}^2 \times \text{height}$	40 cu ft
Order volume	+ 5–10% waste for uneven subgrade and spillage	≈ 1.63 cu yd
In bags	80 lb bag ≈ 0.6 cu ft (one yard ≈ 45 bags)	≈ 74 bags

THE CONSTANT

27 cubic feet = 1 cubic yard. Convert cu ft ÷ 27, then add waste. Ready-mix trucks bill by the yard — compare their price against the bag route before you commit.

04 Mulch: area × depth ÷ 324

STEP	RULE	250 SQ FT BED, 3 IN DEEP
Volume	area (sq ft) × depth (in) ÷ 324	2.31 cu yd
In bags	yards × 27 ÷ bag size, rounded up	≈ 32 bags (2 cu ft)
Depth	2–3 inches for beds; never pile against trunks	—

BULK VS BAGGED

Bulk mulch runs \$35–\$60 per cubic yard delivered in most markets. Past roughly 3 cubic yards, bulk usually wins on price — but bagged wins on convenience for small beds and no-driveway deliveries.

05 Five rules that save money

Measure twice, order once. Write every dimension down before shopping; break odd rooms into rectangles and add them up.

Round up, never down. Paint gallons, flooring boxes, mulch bags, concrete yards — returning leftovers is easier than a second trip for one missing unit.

Trust the label over the rule. Coverage and yields printed on the product beat any cheat sheet when they disagree.

Add waste the honest way: 5–10% covers cuts, mistakes, and subgrade surprises on nearly every project.

Price the whole job, not just materials — delivery fees, underlayment, tape, and hardware are where budgets leak.

RUN THE NUMBERS

Free calculators at calcforaged.com — Paint · Flooring · Concrete · Mulch · plus 20 more for the kitchen, the wallet, and the wedding.